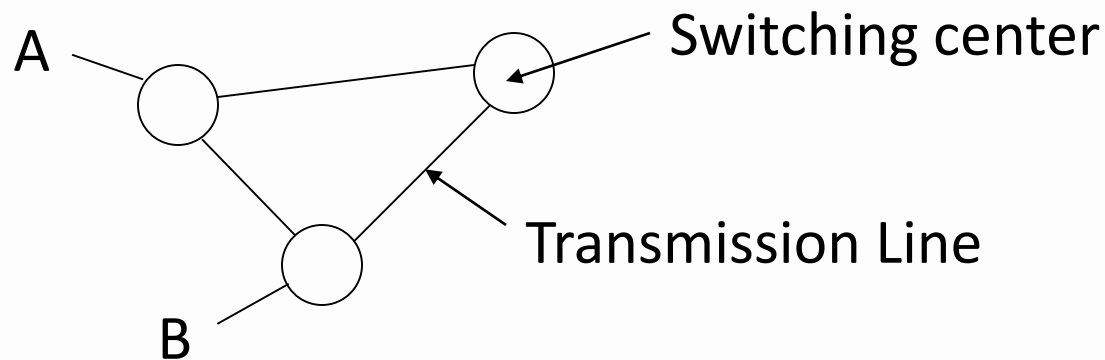
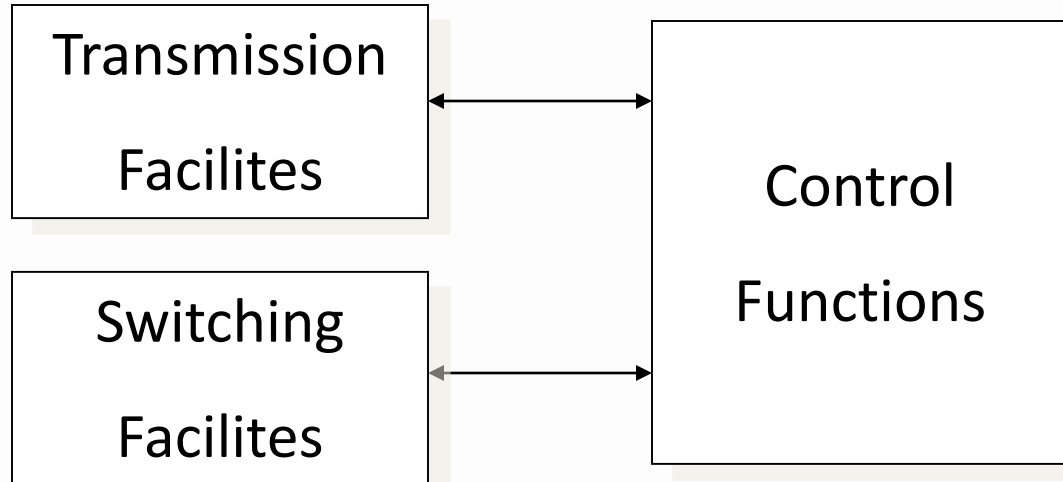
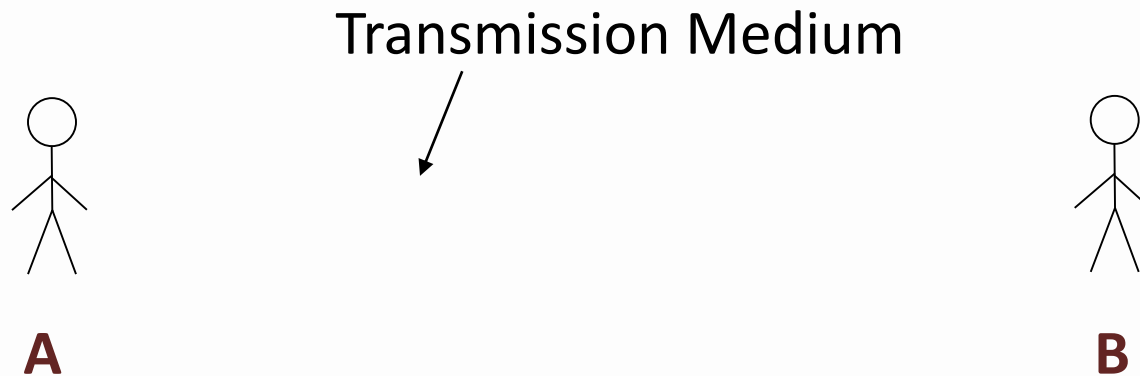


# Switching

# Roles of Switching and Transmission



# Roles of Switching and Transmission

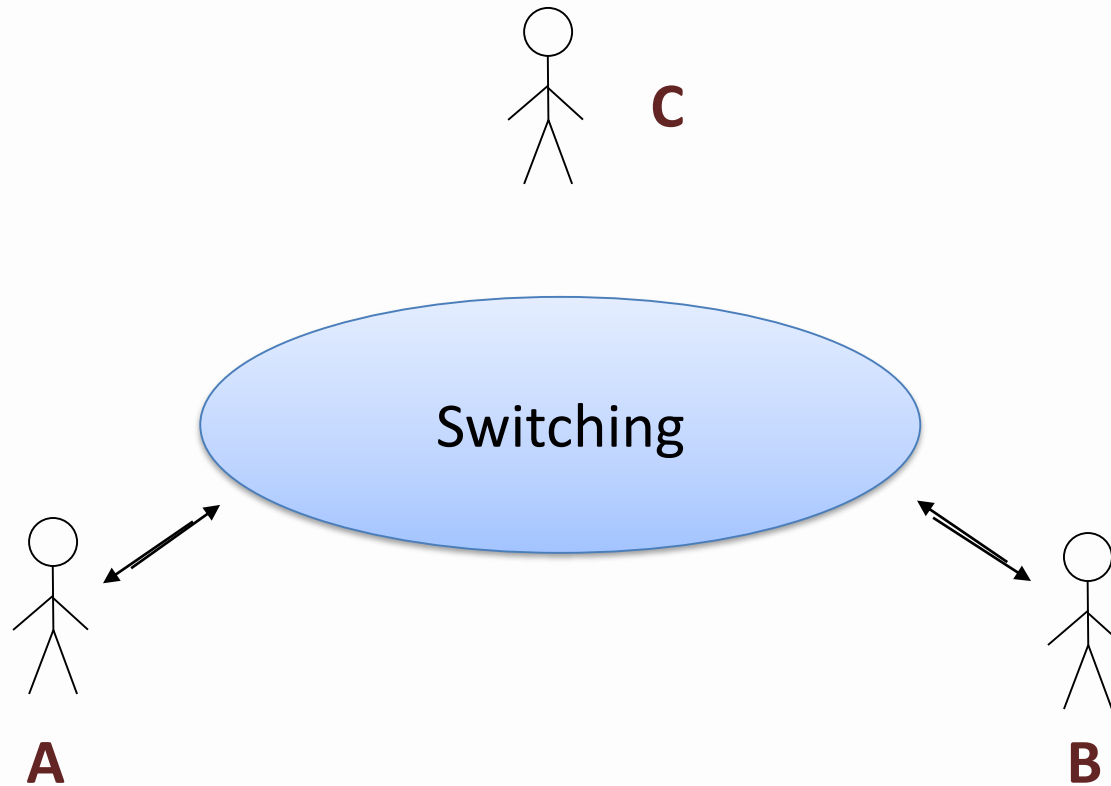


When there are only two users, information from A is by default destined for B, and vice versa

**For a two-user network, switching is not required**

**Fig. 1.1. A two-user network; switching is not required**

# Roles of Switching and Transmission

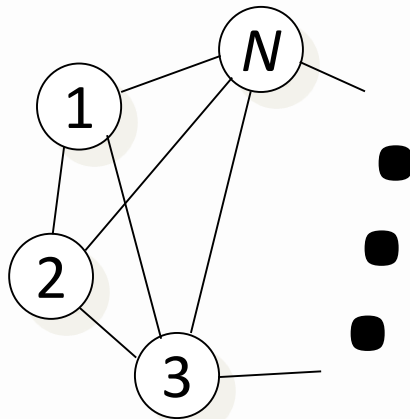


Information from A may be destined for B or C

**For a three-user network, switching is required**

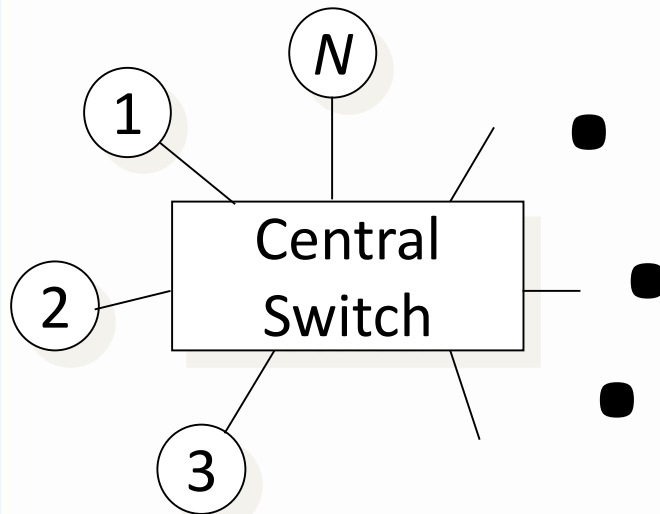
**Fig. 1.2. A three-user network; switching is required**

# Roles of Switching and Transmission



Switching is performed at user's location by selecting one of the links for reception

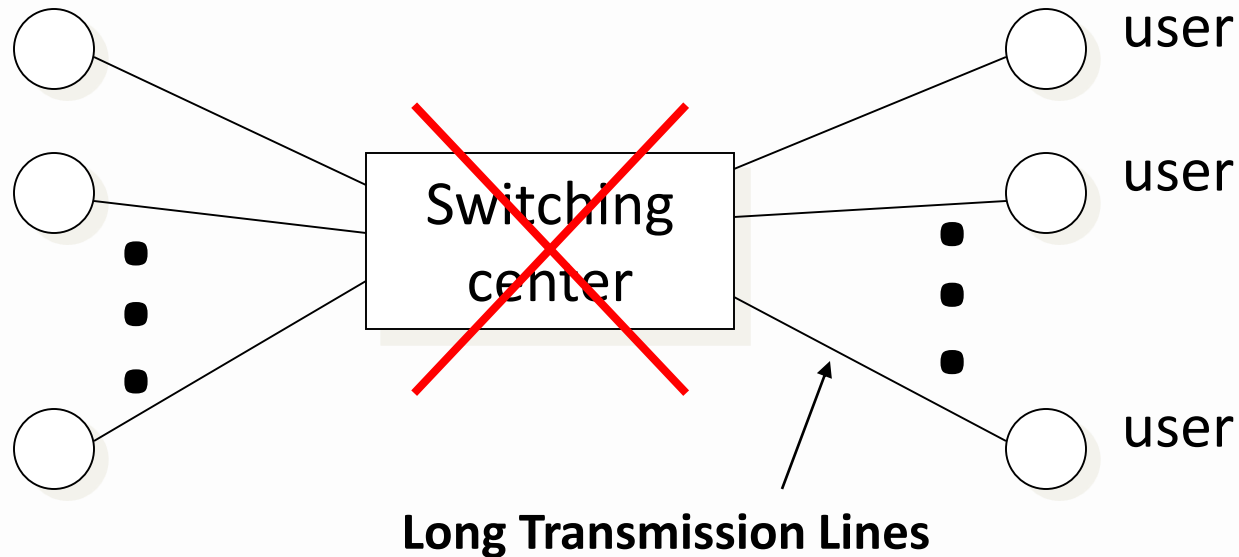
$$\# \text{ of bi-directional links} = N(N-1)/2$$



Switching is performed by a central switch

**Fig. 1.3. N-user networks with switching performed  
(a) at users' location (b) by a central switch**

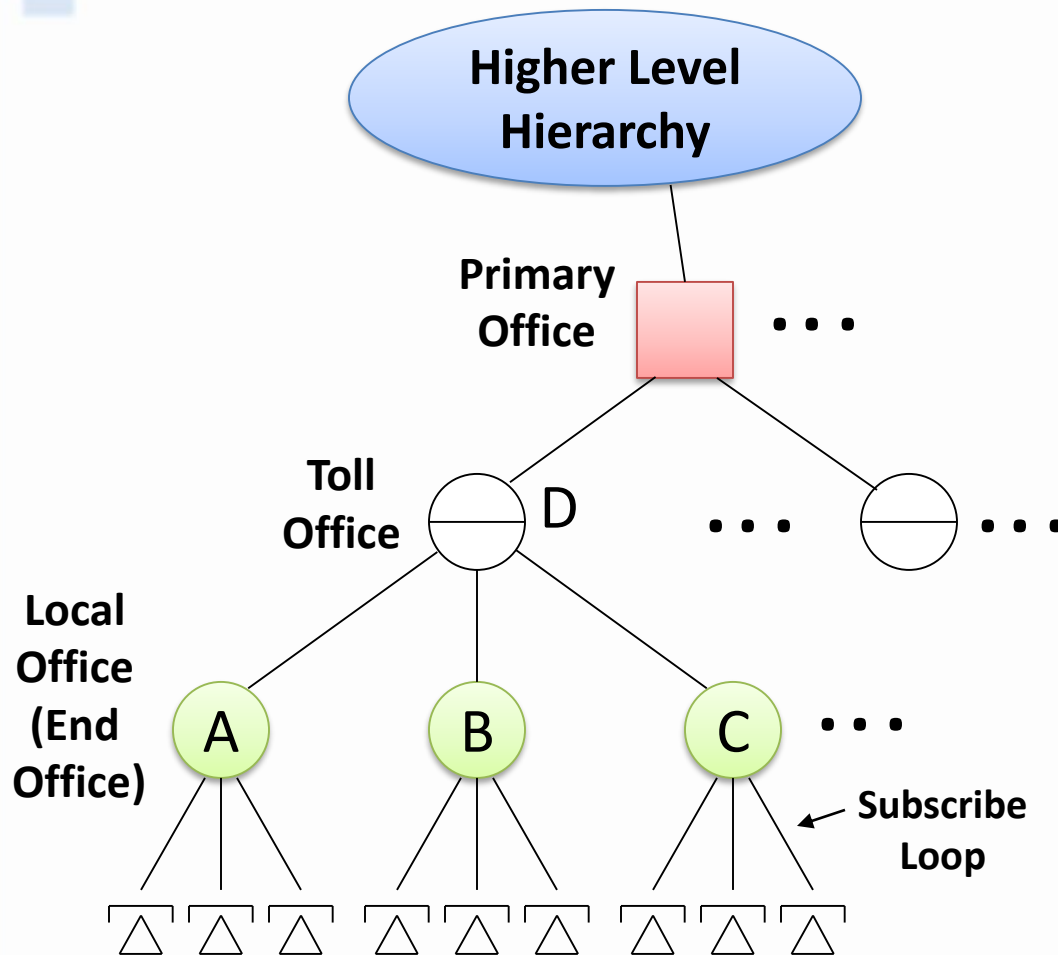
# Roles of Switching and Transmission



In case the totally centralized switch is not attractive, no communication can be processed between any users

**An example in which totally centralized switching is not attractive**

# Transmission Hierarchy



i.) Decrease of Traffic volume as going up in hierarchy

ii.) Resources at upper hierarchy shared by larger population

iii.) Hierarchical routing

Fig. 1.4. Telephone network hierarchy

# Multiplexing and Concentration

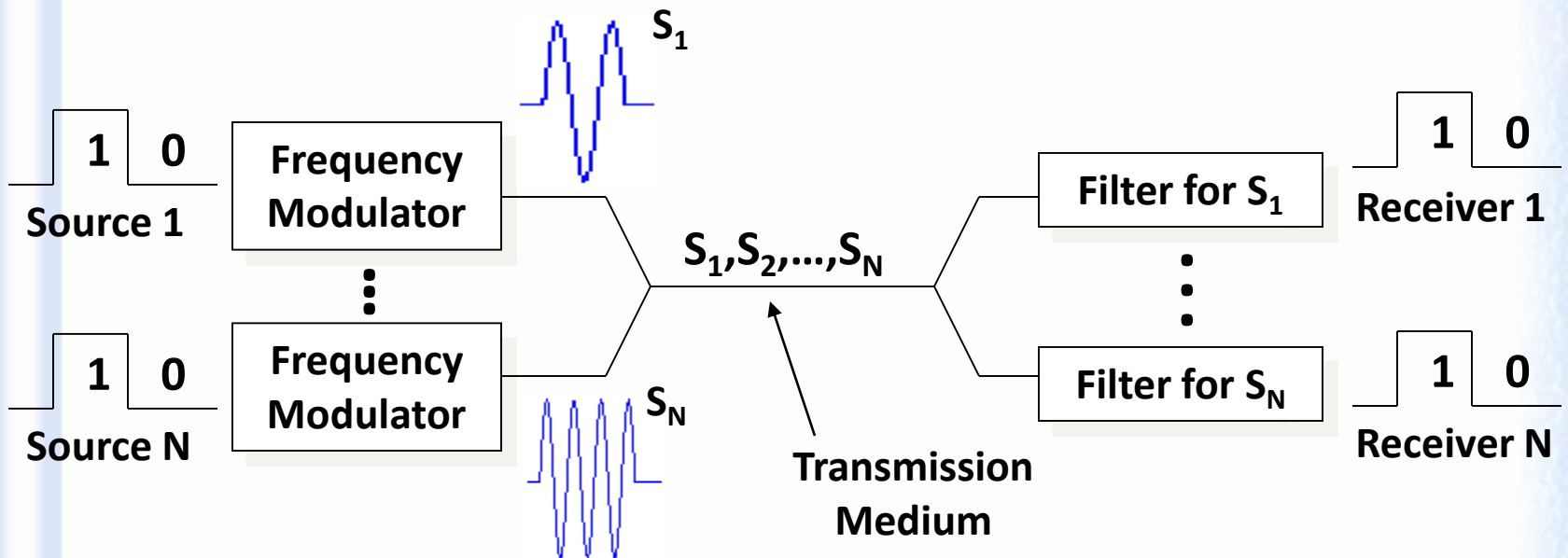


Fig. 1.5. Frequency-division multiplexing



# Multiplexing and Concentration

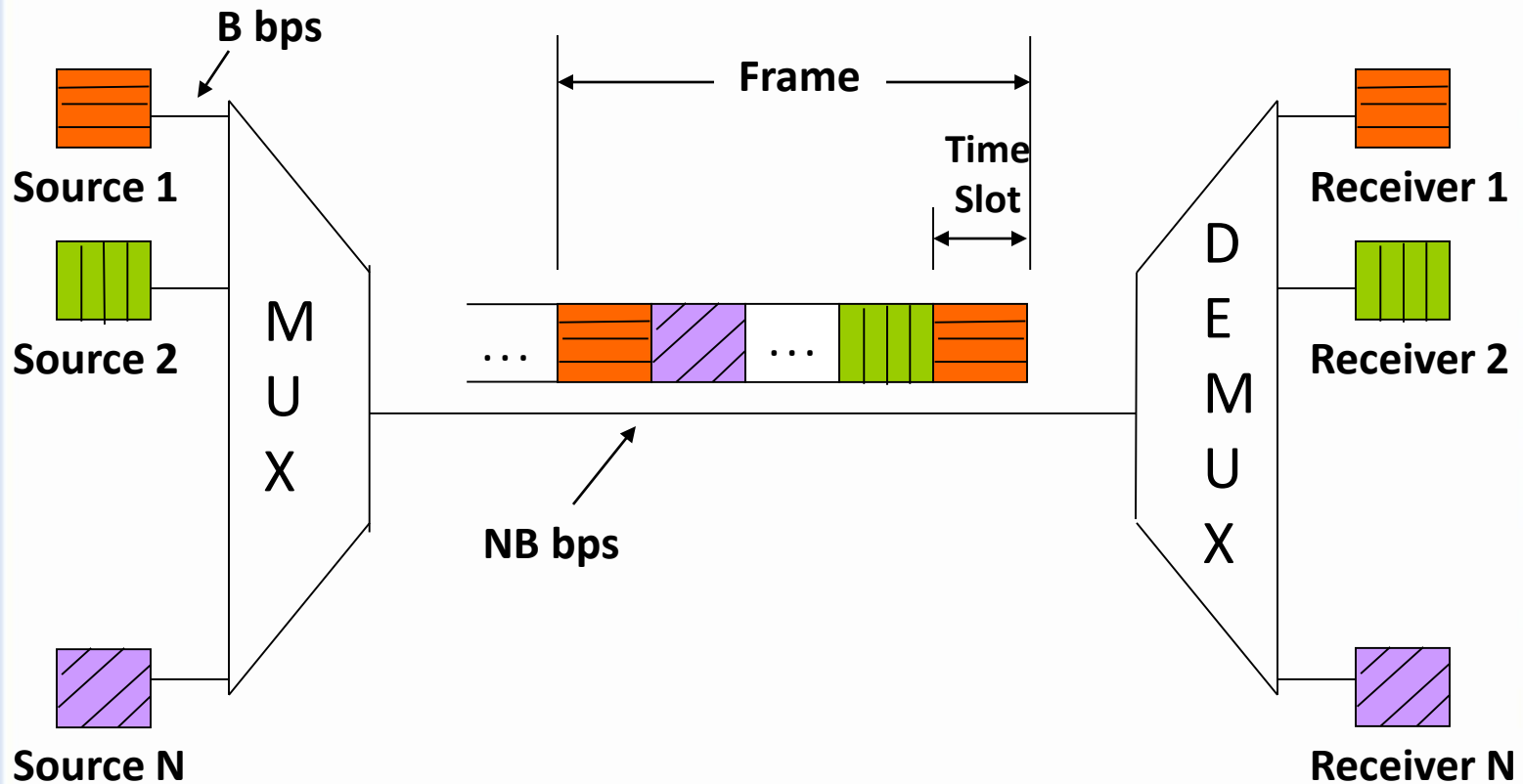
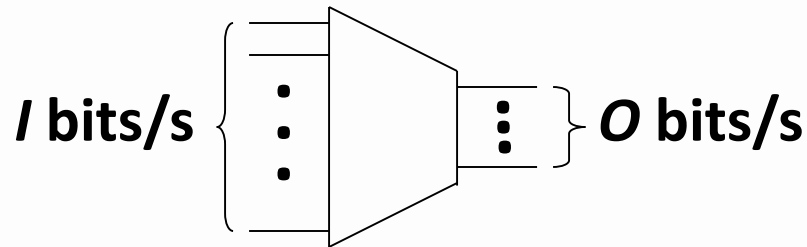


Fig. 1.6. Time-division multiplexing

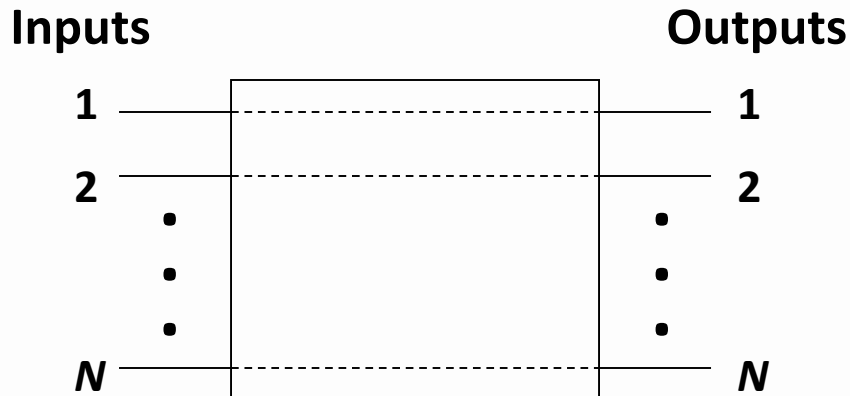
# Multiplexing and Concentration

## Difference between Multiplexing and concentration :



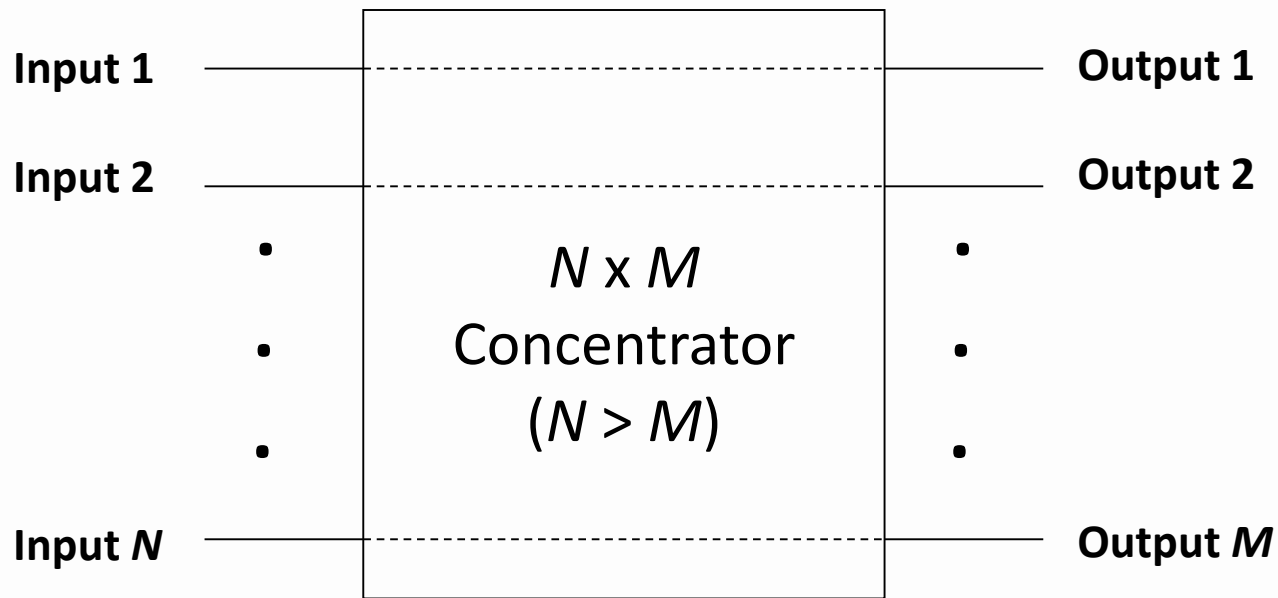
$I = O$  in mux. system  
 $I > O$  in concentrator

## Difference between Multiplexing and Switching :



Connectivity is static  
in mux, system but  
dynamic in switching  
system

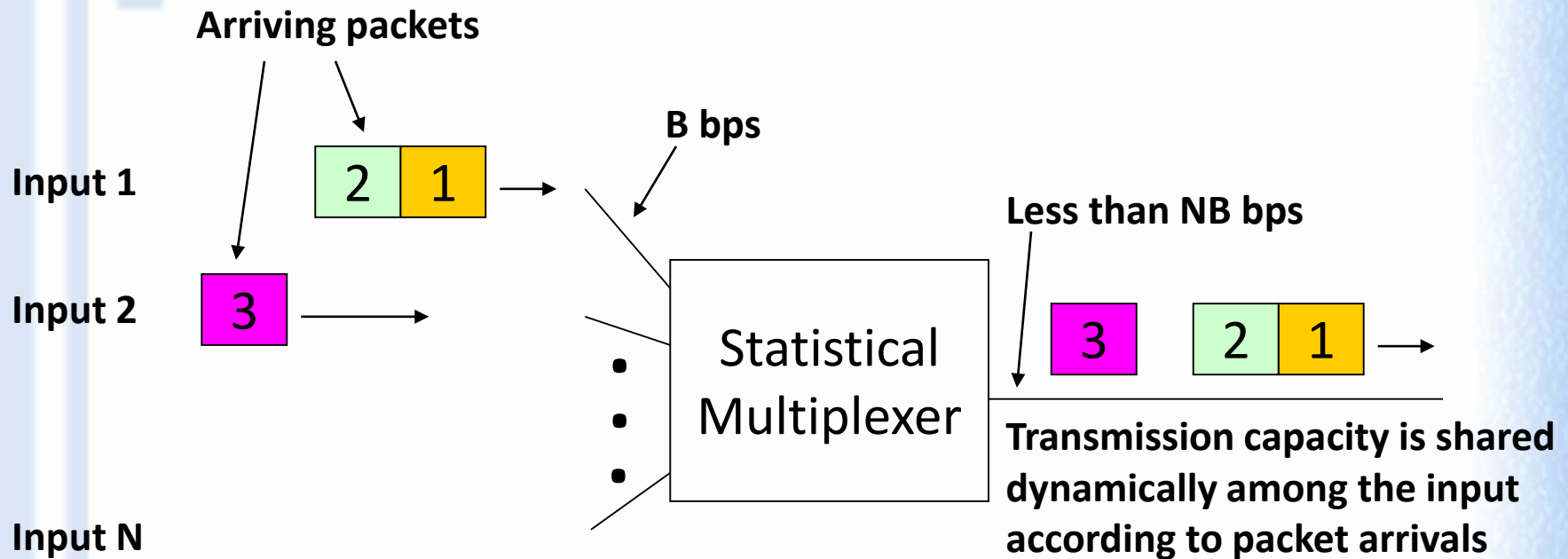
# Multiplexing and Concentration



An active input is assigned to one of the outputs. It does not matter which output is assigned.

**Fig. 1.7. An  $N \times M$  concentrator**

# Multiplexing and Concentration



- Combines multiplexing and concentration
- Packet header indicating destination
- Switching involved
- Connectivity changes with time

**Fig. 1.8. A Statistical multiplexer**

# Multiplexing and Concentration

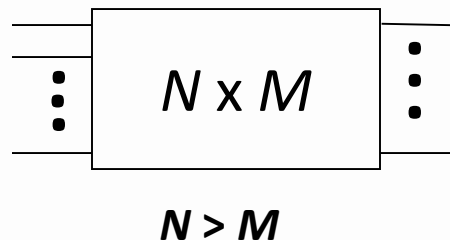
## Sessions, circuits (concentration) :

- Session needs to be set up before communication
- Exclusive and non-exclusive dedication of resources during connection
- Virtual circuits (not connection oriented)

## Messages (Statistical multiplexing) :

- variable packets
- talk spurts

TASI :



Satellite Links assigned  
based on talkspurts

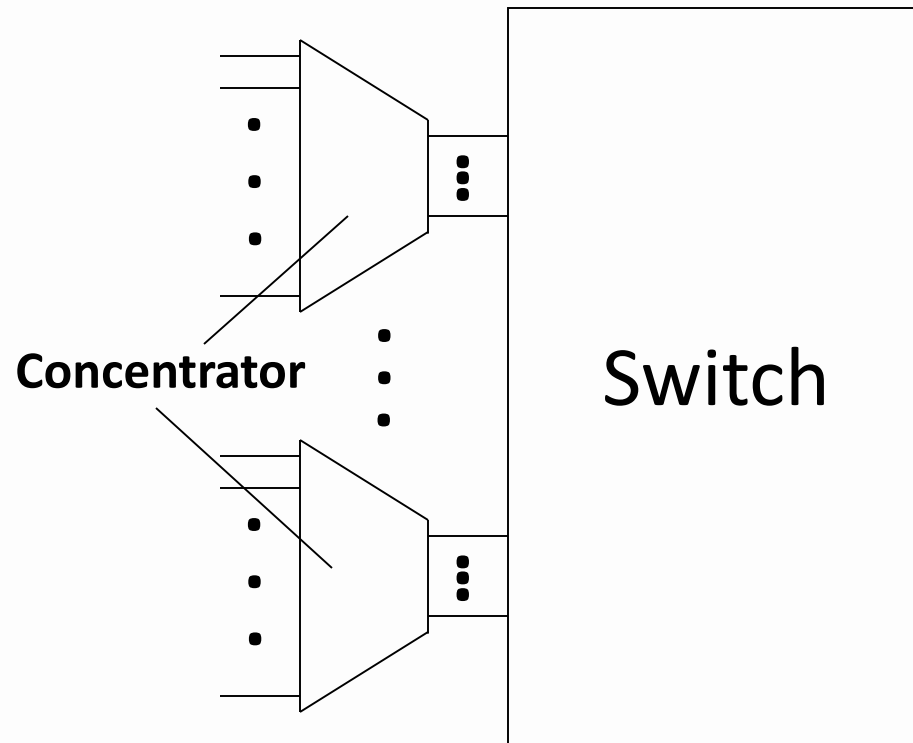
# Timescales of Information Transfer

## Packets and Cells :

- Transport Entities
- Variable versus fixed length
- Long versus short packets

## Messages are broken into smaller packets in network since:

- Network does not support large packet size
- Most networks are store-and-forward network, less delay under light load
- Prevent a long packet from hogging a communication channel

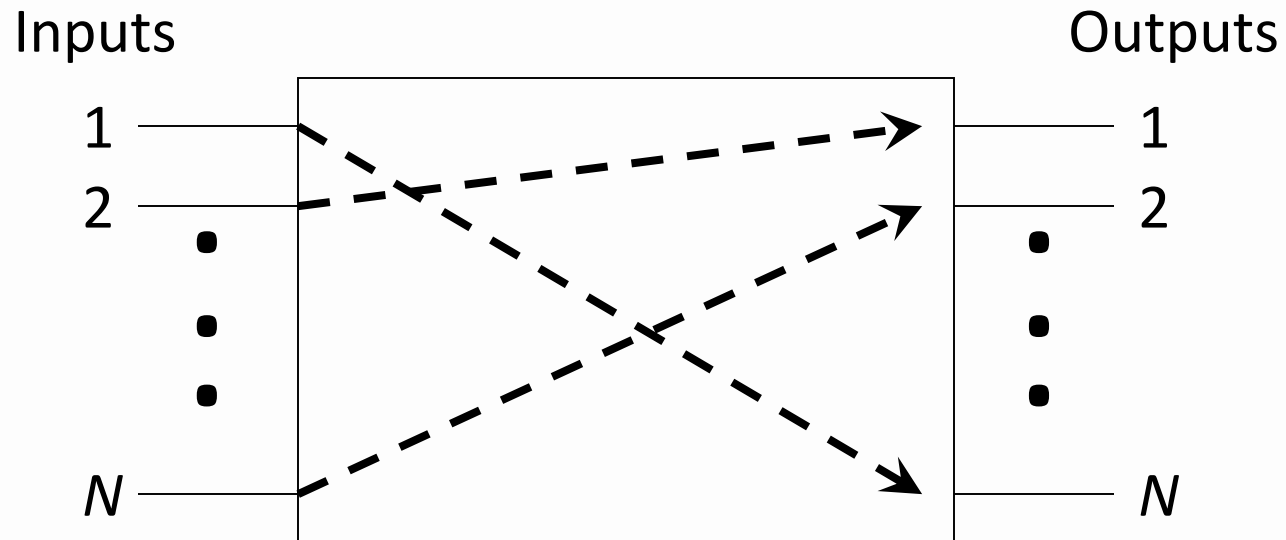


**Concentration used to reduce switching cost**

# Circuit Switch Design Principles

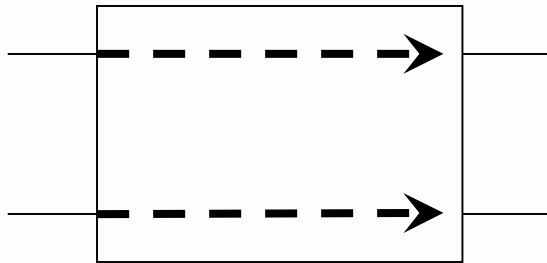


# Space-Domain Circuit Switching

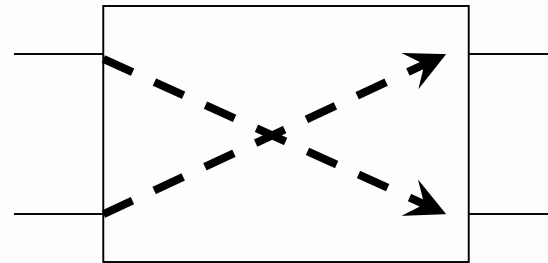


**Fig. 2.1. An  $N \times N$  switch used to interconnect  $N$  inputs and  $N$  outputs**

# Strictly Nonblocking



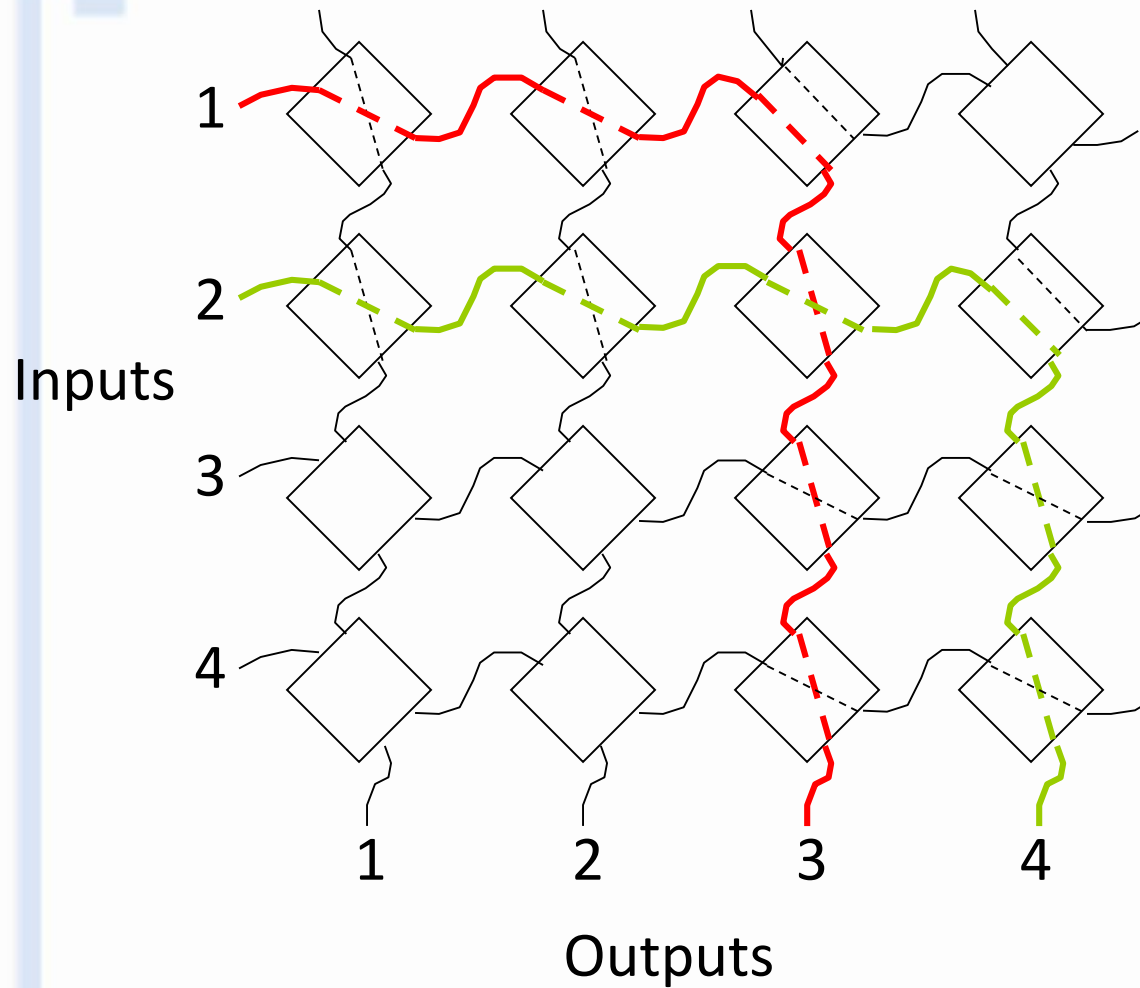
Bar State



Cross State

**Fig. 2.2. Bar and cross states of 2 x 2 switching elements**

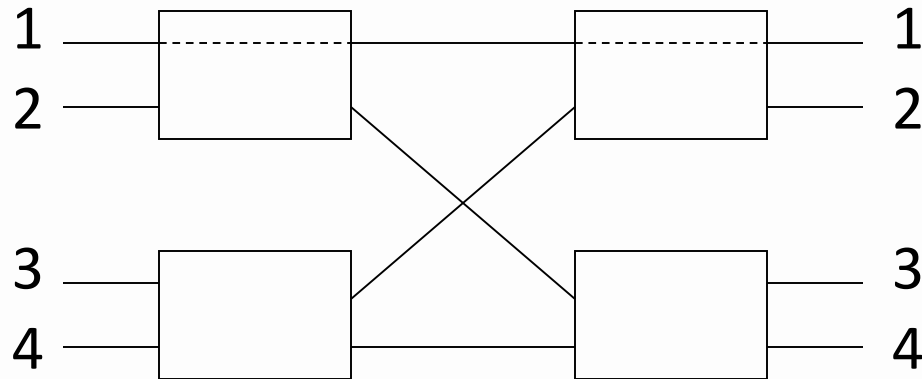
# Strictly Nonblocking



**Connections:**  
Input 1 to Output 3  
Input 2 to Output 4

**Fig. 2.3. (a) Crossbar switch**

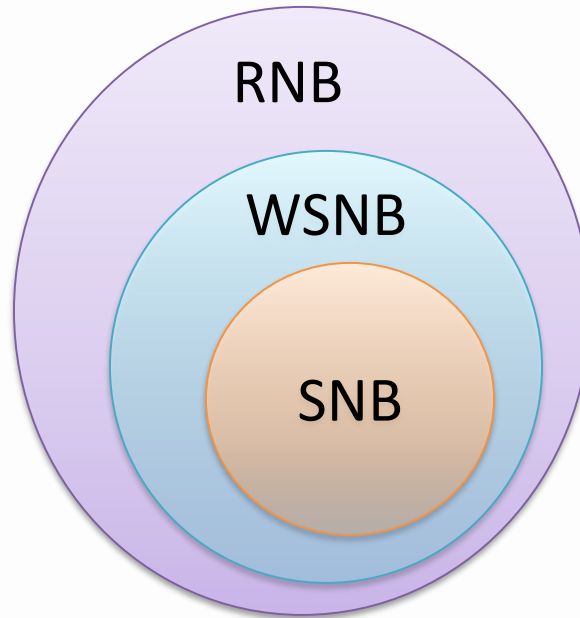
# Blocking



Blocking: Input 2 cannot be connected to output 2  
if input 1 is already connected to output 1

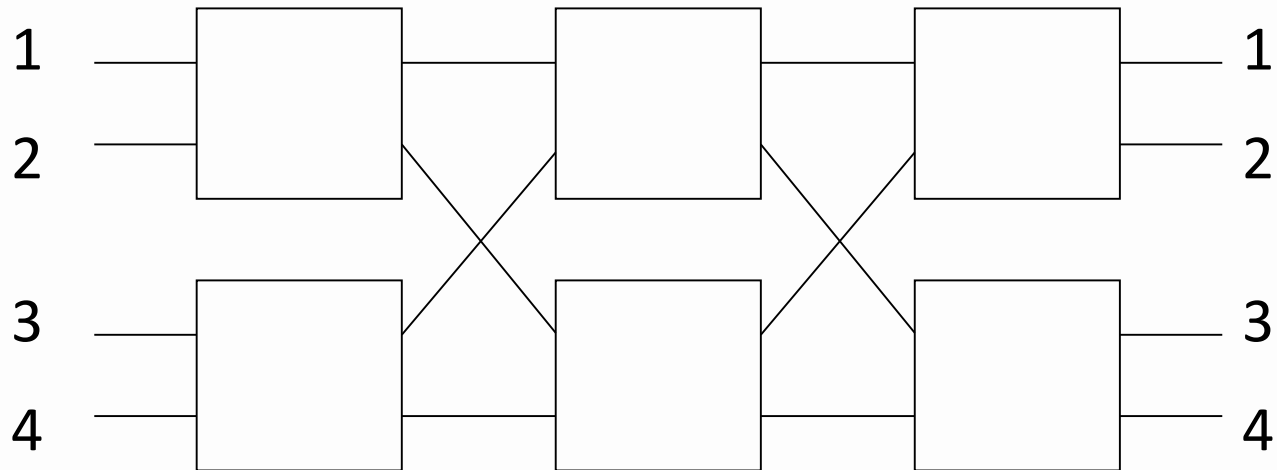
**Fig. 2.3. (b) banyan switch**

# Nonblocking Properties



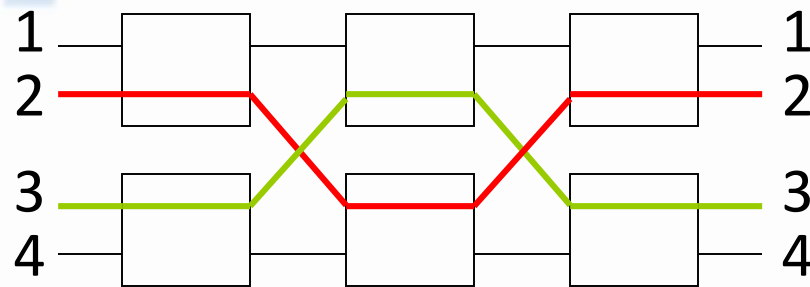
- RNB** — Rearrangeably Nonblocking
- WSNB** — Wide-sense Nonblocking
- SNB** — Strictly Nonblocking

# Rearrangeably Nonblocking



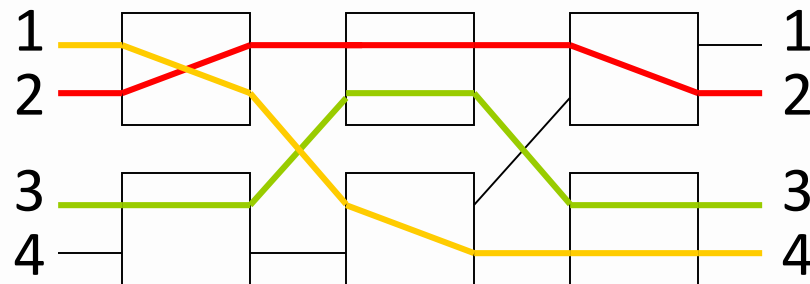
**Fig. 2.4. (a) A 4 x 4 rearrangeably nonblocking switch**

## Rearrangements



Connection cannot be set up between input 4 and output 1

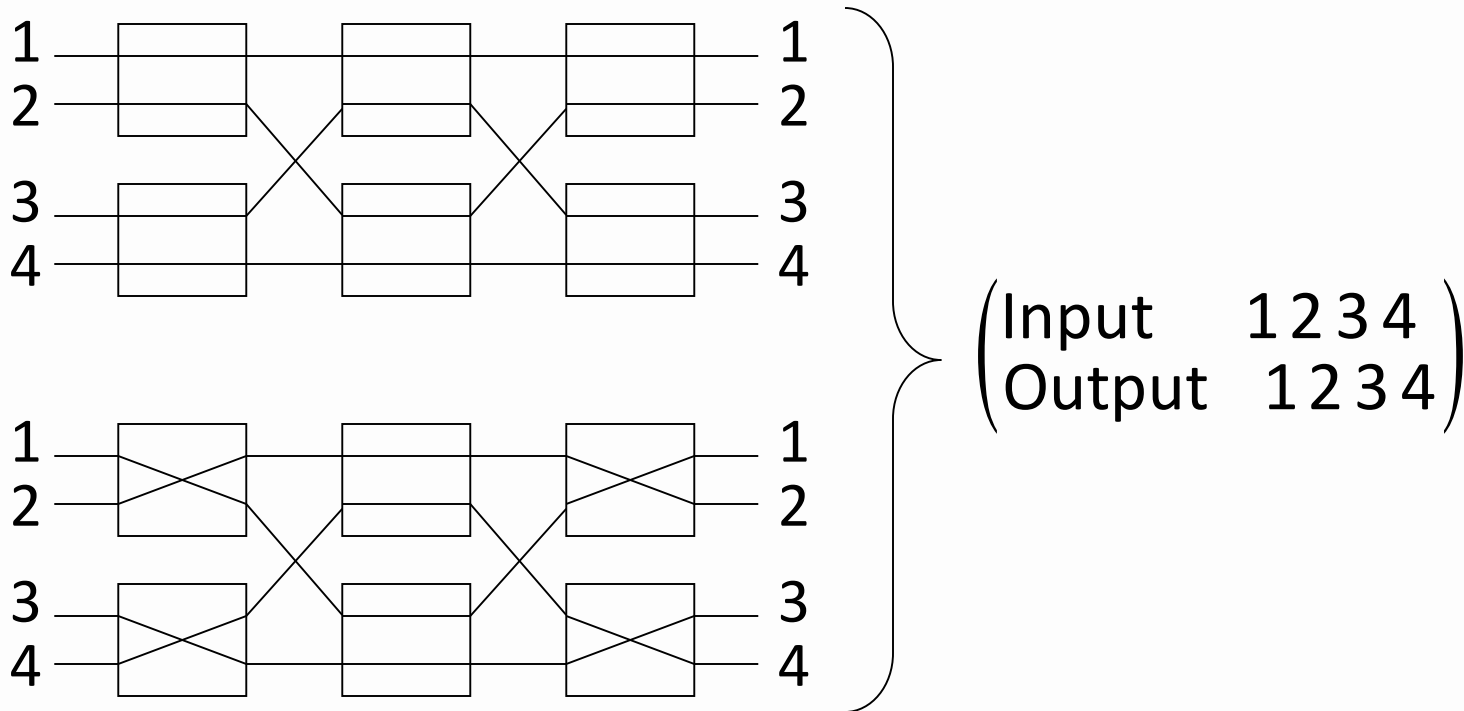
**Fig. 2.4. (b) a connection request from input 4 to output 1 is blocked**



Connection can now be set up between input 4 and output 1

**Fig. 2.4. (c) Same connection request can be satisfied by rearranging the existing connection from input 2 to output 2**

## Two states corresponding to the same mapping :

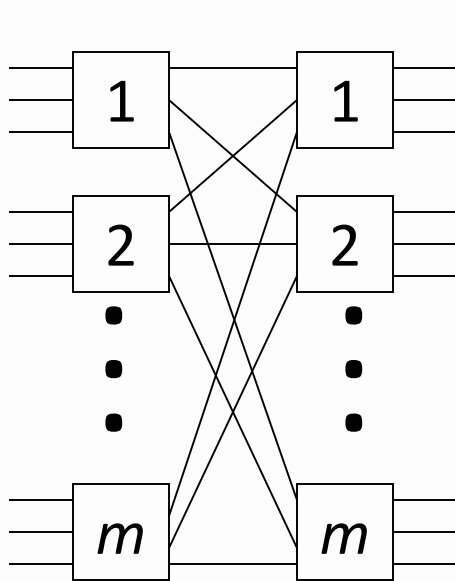




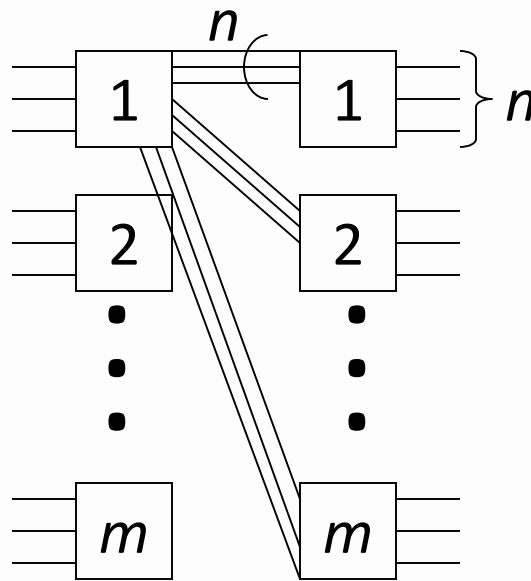
## Complexity of nonblocking switches :

How to build large switch from smaller switches?

Problems with two-stage networks :



(a)

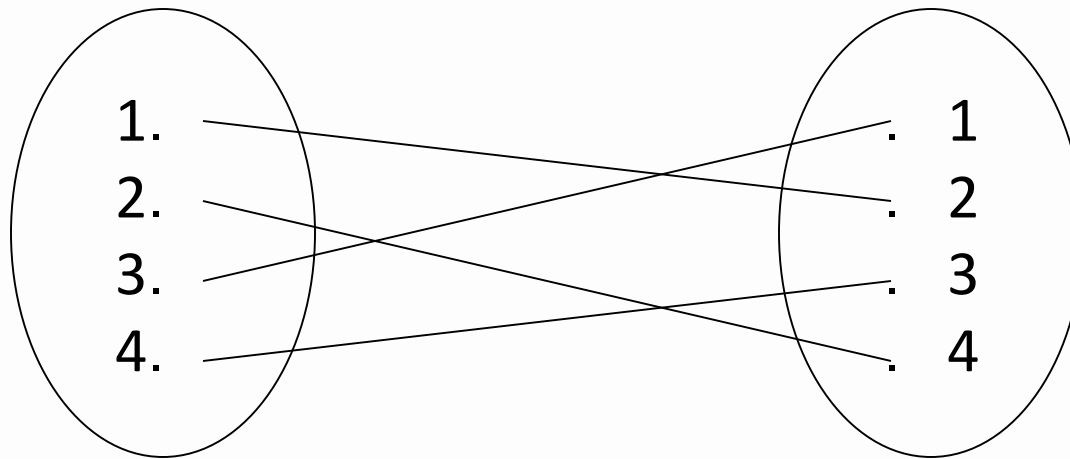


(b)

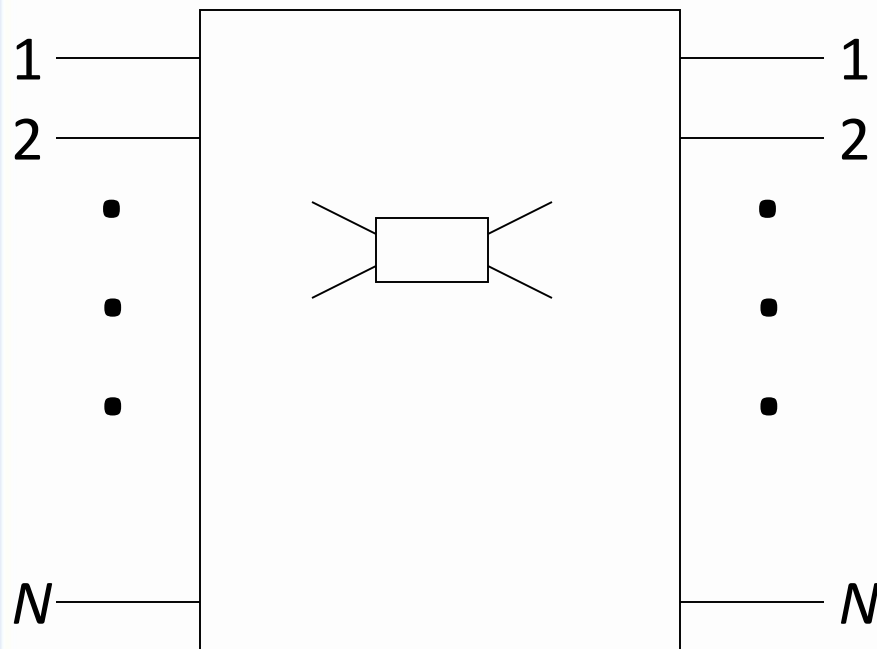
$$N = mn$$

$$\begin{aligned}\text{\# lines} &= m^2n \\ &= mN\end{aligned}$$

Bandwidth expansion  
factor =  $m$



**Fig. 2.5. (a) An example of one-to-one mapping from input to output**



$N!$  mappings  
 $M$  crosspoints

# states  $\geq$  # mappings

$$2^M \geq N!$$

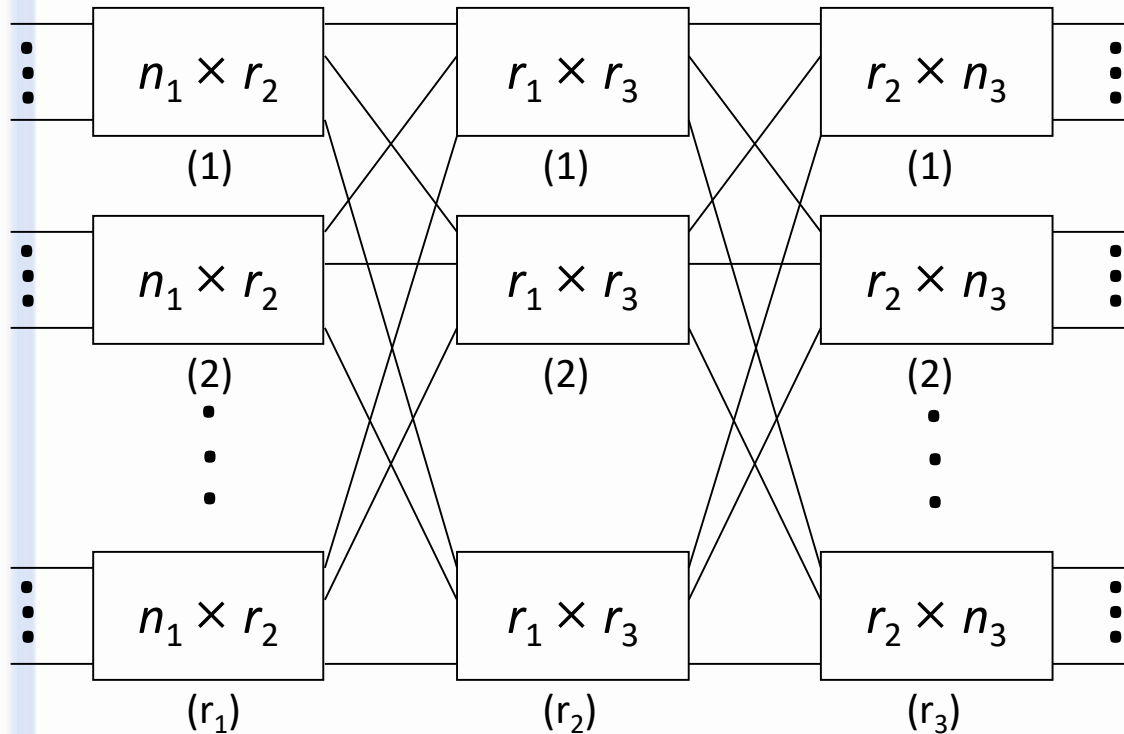
$$M \geq \log_2 N!$$

$$\approx N \log_2 N$$

for large  $N$

**Fig. 2.5. (b) Number of crosspoints needed for nonblocking switch**

# Clos Switching Network



$$n_1 r_1 = n_3 r_3 = N \text{ for } N \times N \text{ switch}$$

$r_i$  — # switch modules in column  $i$

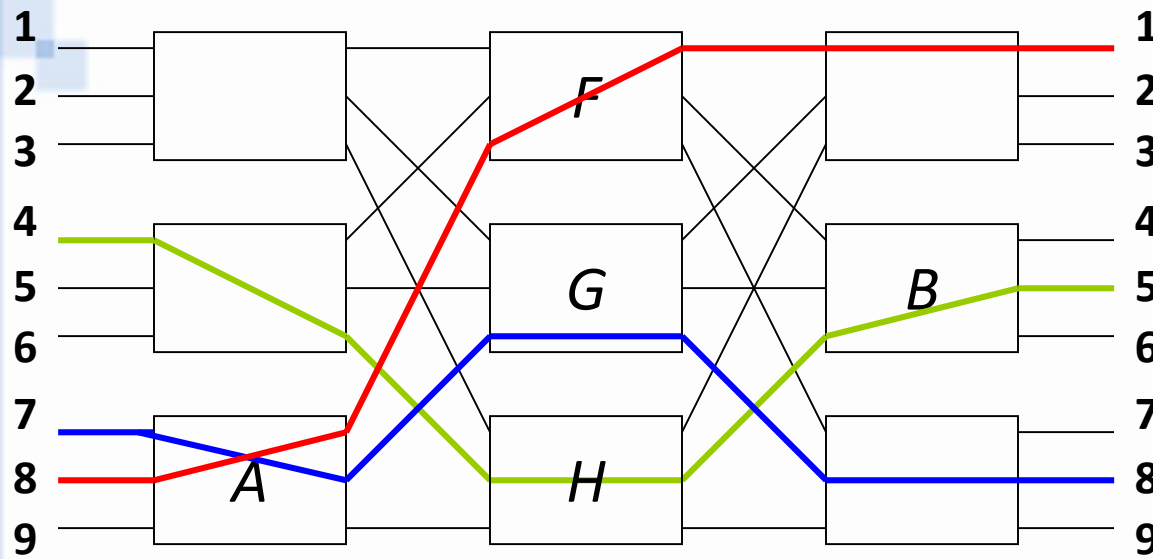
$n_1$  — # inputs in column 1 module

$n_3$  — # outputs in column 3 module

Necessary condition for nonblocking:

$$r_2 \geq n_1, n_3$$

Fig. 2.6. A three-stage clos switch architecture



**Key:**

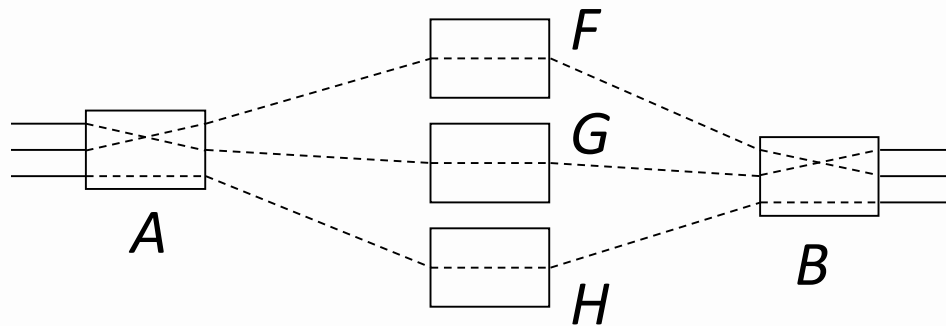
Find a commonly accessible middle node from both input and output nodes

A request for connection from input 9 to output 4 is blocked

$S_A$  = middle-stage nodes used by A  
 $= \{ F, G \}$

$S_B$  = middle-stage nodes used by B  
 $= \{ H \}$

**Fig. 2.7. An example of blocking in a three-stage switch**

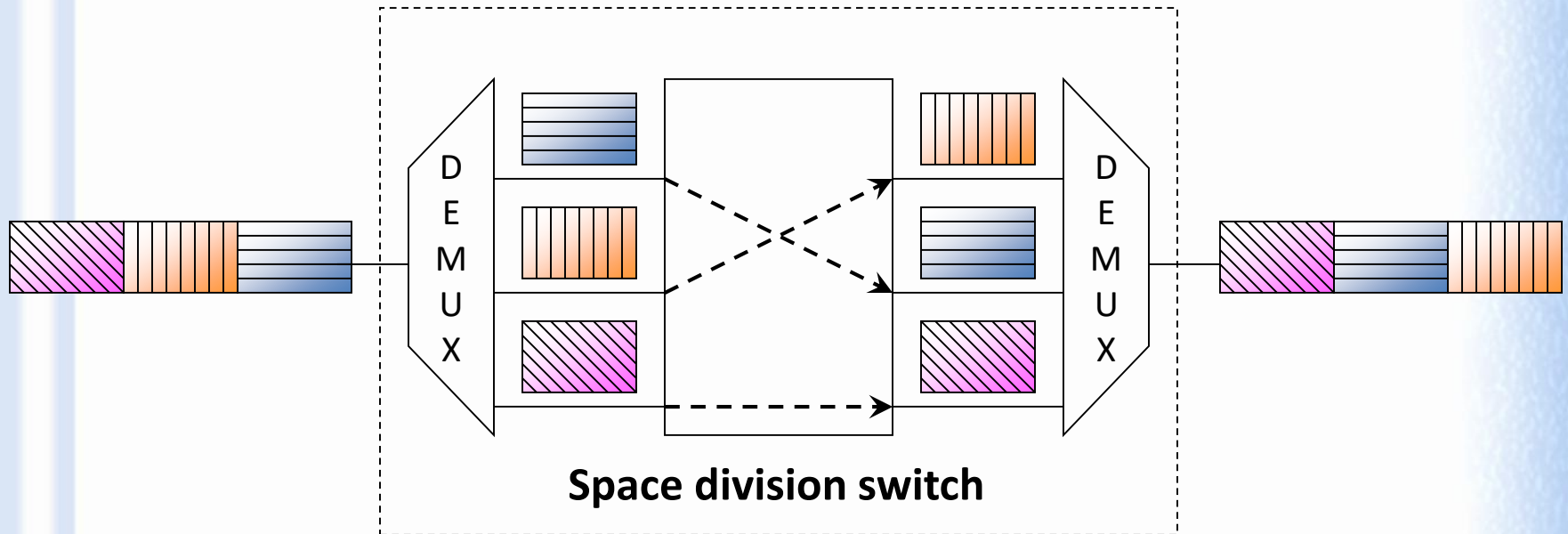


**Stage 3 switch**

|                                   | 1 | 2 | <i>B</i> |                |  | $r_2$ |
|-----------------------------------|---|---|----------|----------------|--|-------|
| 1                                 |   |   |          |                |  |       |
| 2                                 |   |   |          |                |  |       |
| <b>Stage 1 switch</b><br><i>A</i> |   |   |          | <i>F, G, H</i> |  |       |
|                                   |   |   |          |                |  |       |
|                                   |   |   |          |                |  |       |
|                                   |   |   |          |                |  |       |
| $r_1$                             |   |   |          |                |  |       |

**Fig. 2.8. The connection matrix of the three-stage network**

# Time-Domain Switching



**Fig. 2.9. Performing time-slot interchange using space-division switch**

# Time-Domain Switching

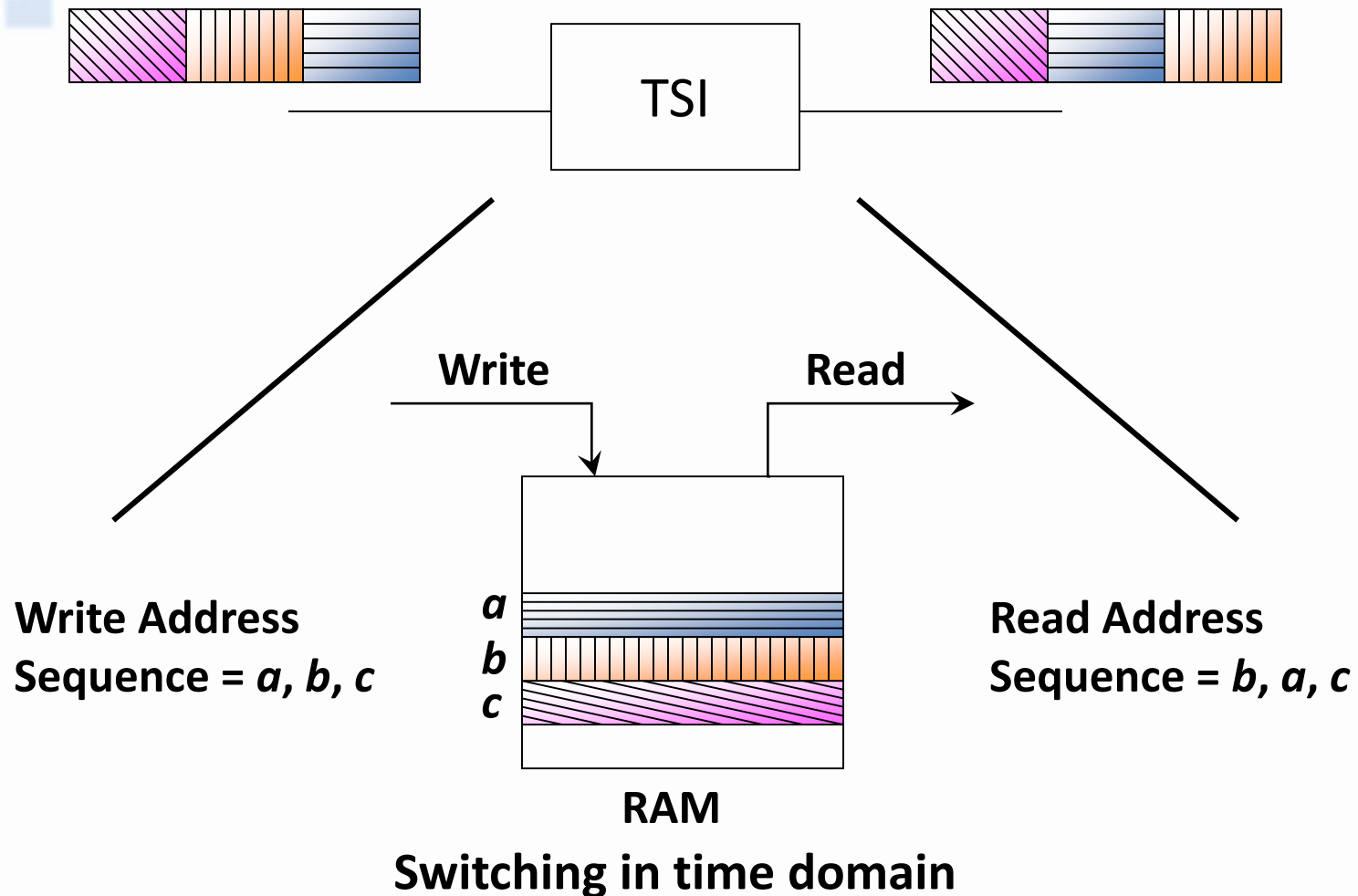


Fig. 2.10. Direct time slot interchange using random access memory (switching in the time domain)



## Example for Time-Domain Switching

Example: memory access time

$$T-1 \text{ rate} = 1.5 \text{ Mbps}$$

$$N = 24$$

Each data source is 64 kbps

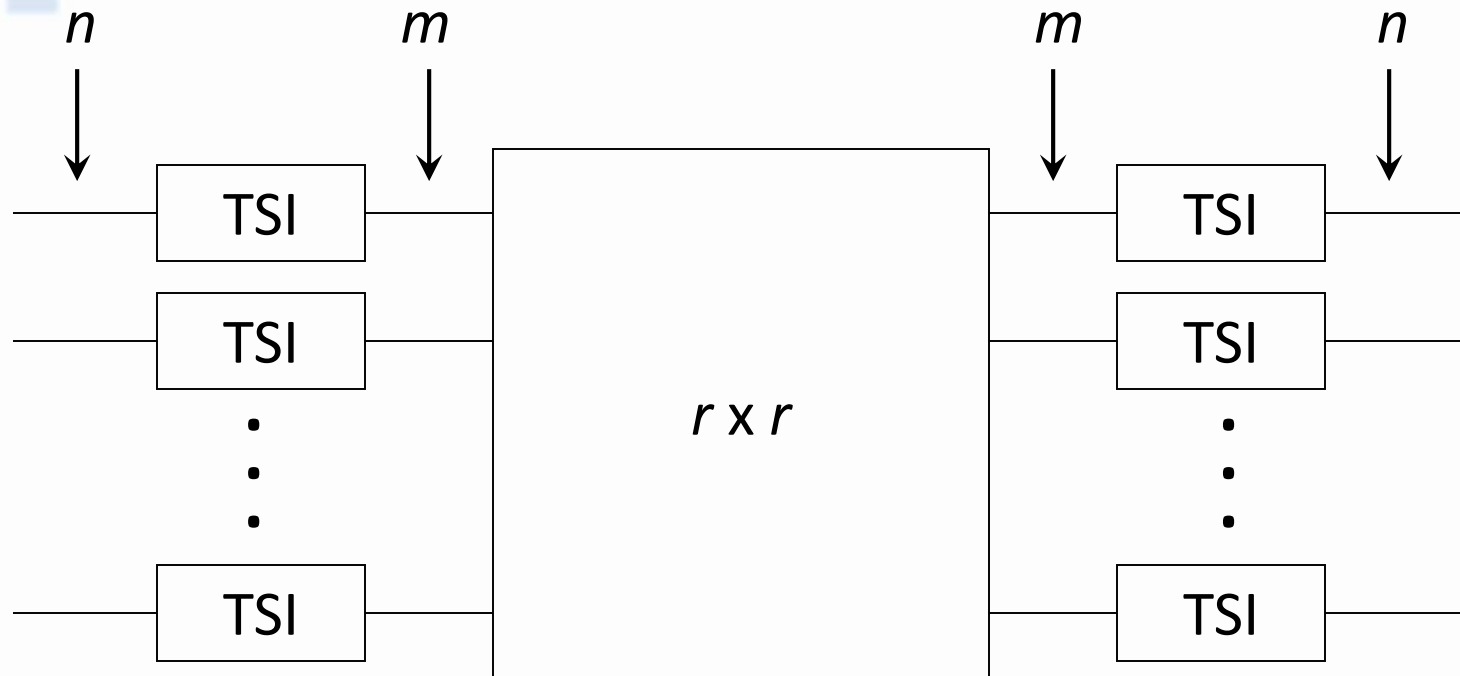
One byte per time-slot

$$\begin{aligned} \text{Arrival rate} &= \frac{24 \times 64,000 \text{ bps}}{8 \text{ bits/time-slot}} \\ &= 192,000 \text{ time-slots/sec} \end{aligned}$$

A read and a write required per time-slot

$$\Rightarrow \text{memory access time} = \frac{1}{2 \times 192,000} \approx 2.6 \mu\text{s}$$

# Time-Space-Time Switching



$n, m$  : number of time slots per frame at various points

**Fig. 2.11. A time-space-time switch**

# Time-Space-Time Switching

$(i, j)$  — data on input  $i$  at time-slot  $j$

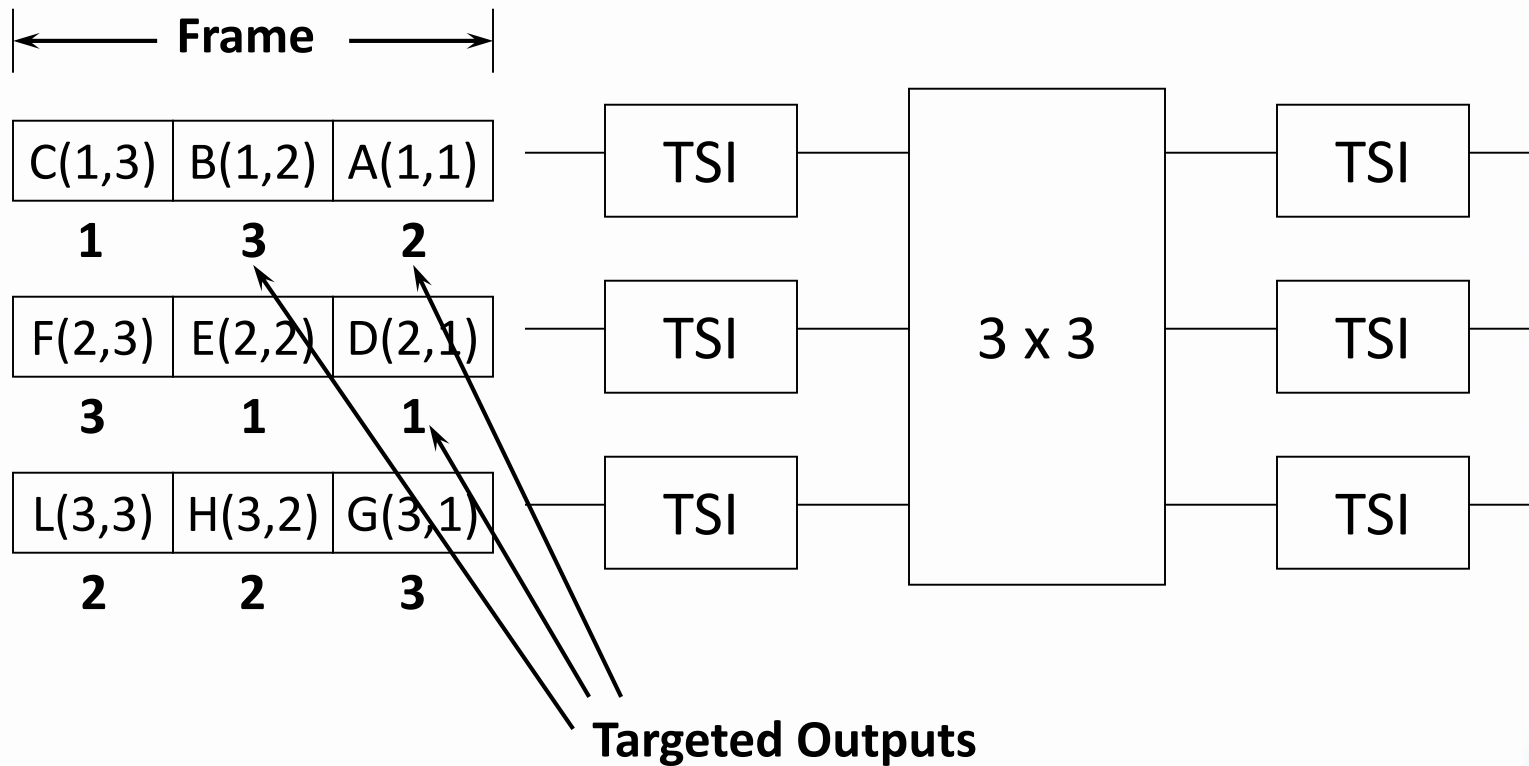
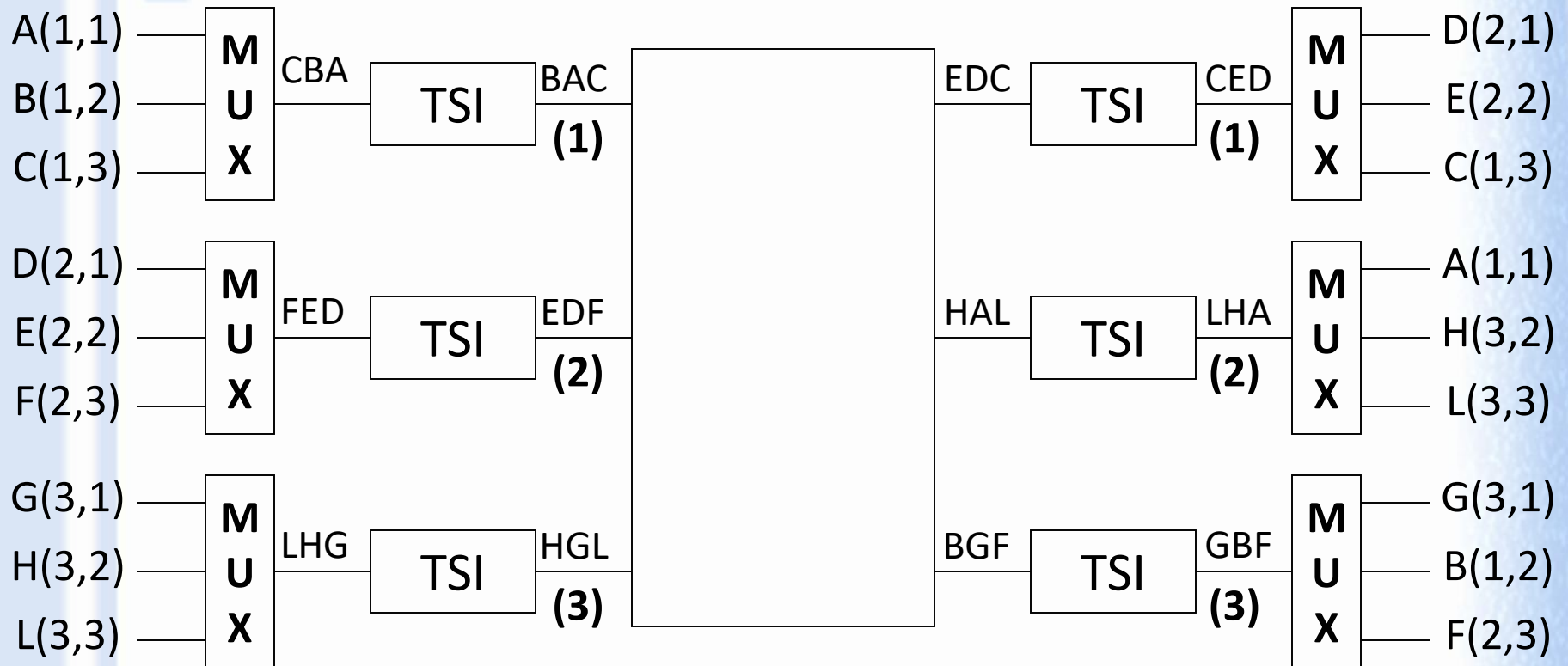
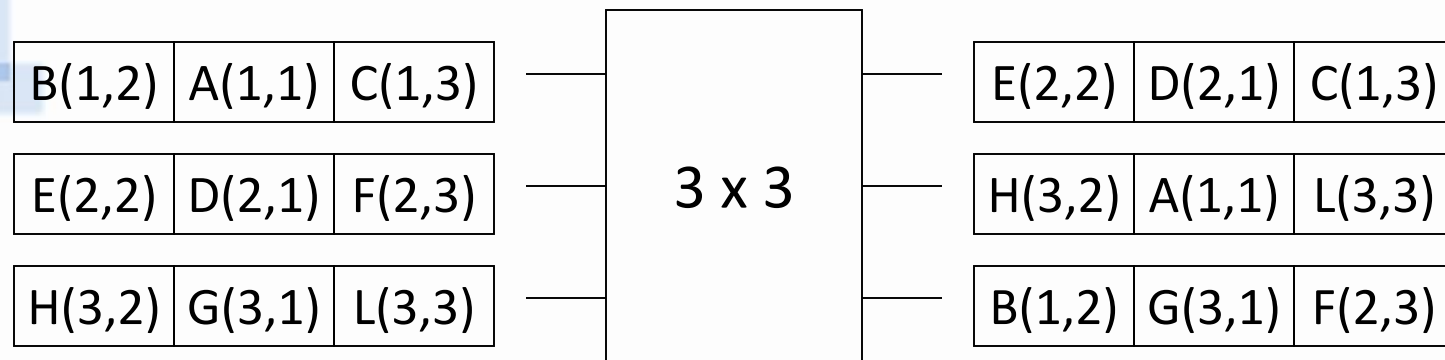


Fig. 2.12 A time-space time switch

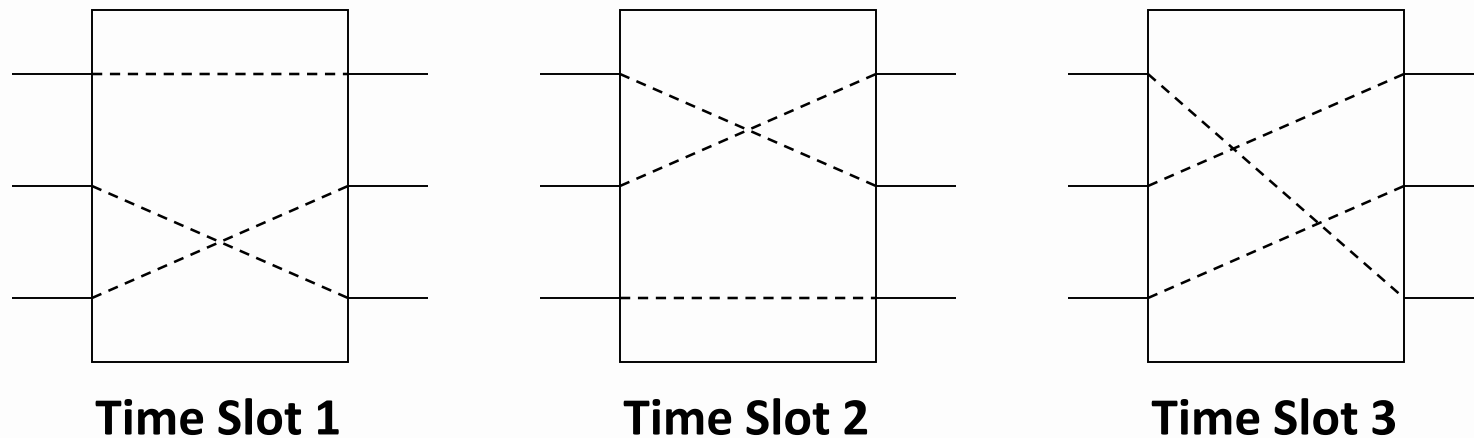
# Time-Space-Time Switching



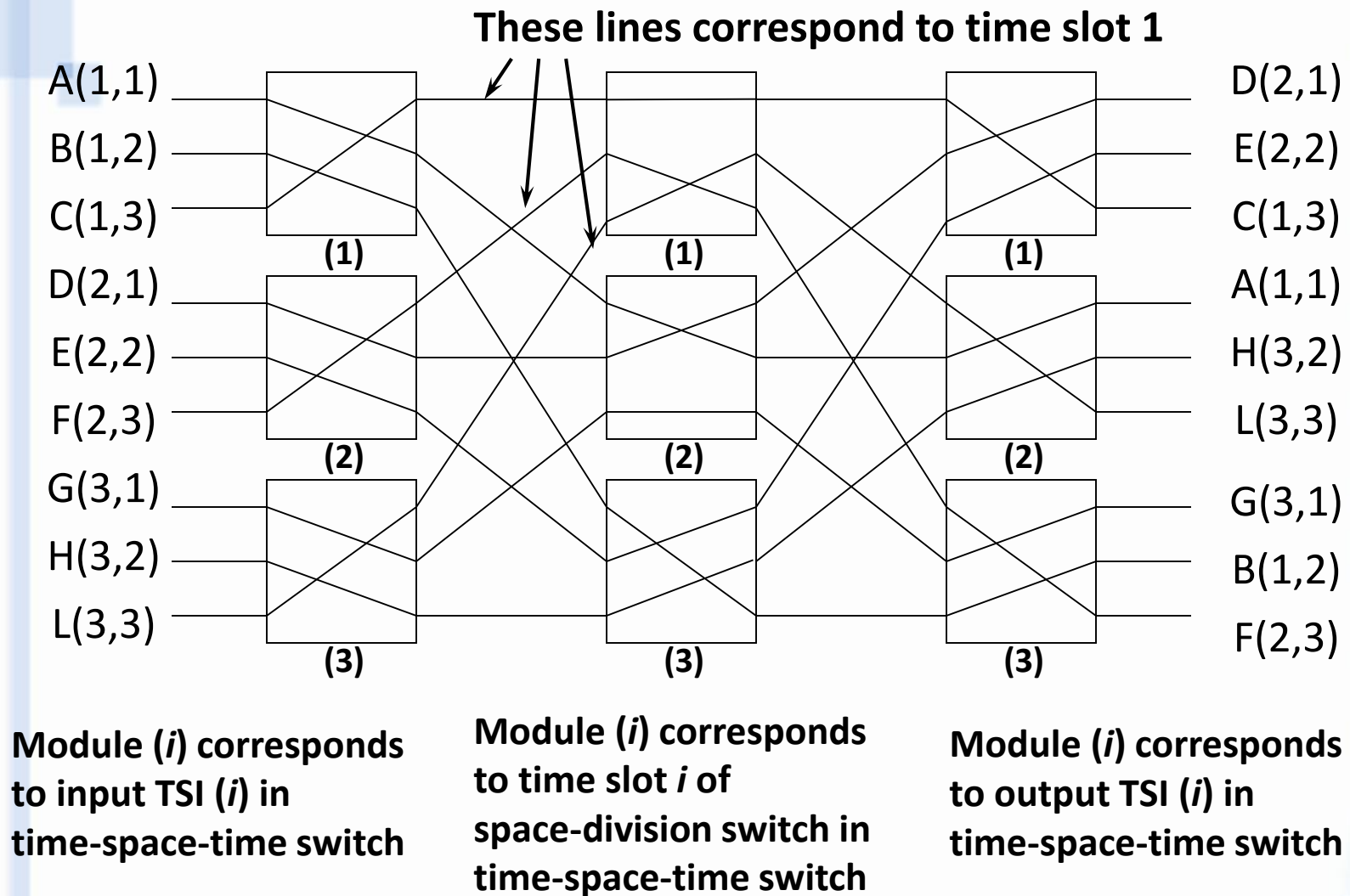
**Fig. 2.13. Equivalent of time-space-time switching and three-stage space switching**



$(i, j)$  — data on input  $i$  at time slot  $j$



**Fig. 2.14. Input-output mapping changes from slot to slot in space-division switch in time-space-time switching**



**Fig. 2.25. Equivalent of time-space-time switching and three-stage space switching**

Thank You.